

Genetic Algorithm Problem Set

Problem 1: Circular Research Collaboration Optimization

Eight senior researchers must sit around a circular strategy table. Each researcher has two preferred collaborators and one strong conflict.

Preferences:

- 1 prefers (4,6), avoids 3
- 2 prefers (5,7), avoids 8
- 3 prefers (1,8), avoids 6
- 4 prefers (1,2), avoids 7
- 5 prefers (2,6), avoids 4
- 6 prefers (1,5), avoids 3
- 7 prefers (2,8), avoids 4
- 8 prefers (3,7), avoids 1

Tasks:

- (a) Encode the seating arrangement as a chromosome and generate four valid parent chromosomes.
- (b) Propose a suitable maximizing fitness function that accounts for weighted preferences and conflicts. Identify the two fittest chromosomes from your generated population.
- (c) Discuss whether standard one-point crossover is appropriate for this encoding. Justify your answer.
- (d) Suggest a meaningful mutation strategy suitable for this permutation-based problem.

Problem 2: Nonlinear Integer Constraint Optimization

Let $X_1, \dots, X_8$ be integers in the range 1–200.

The objective is to satisfy the nonlinear condition:

$$X_1X_2 + X_3^2 = 4(X_4 + X_5) - X_6 + X_7X_8$$

Tasks:

- (a) Create an initial population of four chromosomes such that the last gene in each chromosome is less than 60.
- (b) Define a maximizing fitness function based on the closeness to satisfying the equation. Compute the fitness of each chromosome.
- (c) Select the best three chromosomes and perform single-point crossover at two different indices to generate four offspring.
- (d) Explain what problem may arise if the first four genes in every chromosome remain small (e.g., ≤ 5) without mutation.

Problem 3: Drone Route Optimization

A drone must visit six monitoring stations exactly once and return to the starting station.

Coordinates:

$A(3, 5)$ $B(7, 9)$ $C(12, 4)$ $D(5, 14)$ $E(10, 13)$ $F(15, 7)$

Distance is calculated using Euclidean distance.

Tasks:

(a) Encode the routing problem as chromosomes and generate four parent chromosomes. Define a suitable fitness function and select the two fittest parents.

(b) Apply two-point crossover to generate two offspring.

(c) Propose a mutation strategy appropriate for this routing problem and discuss its impact on fitness.

Problem 4: High-Dimensional Resource Allocation

Ten computational tasks must be assigned priority levels between 1 and 100. The system must satisfy:

$$\sum_{i=1}^5 T_i = 2 \sum_{i=6}^{10} T_i + 50$$

Tasks:

(a) Form a population of four chromosomes satisfying the constraint that $T_{10} < 40$.

(b) Construct an appropriate maximizing fitness function.

(c) Perform crossover operations and describe the generation of offspring.

(d) Explain how premature convergence might occur in this constrained optimization problem.